LECTURE 15
BINARY RESPONSE MODELS

This lecture discusses models in which the dependent variable takes only two values, usually zero or one. For example, Y_i can measure whether or not individual i participates in the labor force, with $Y_i = 1$ (yes) and $Y_i = 0$ (no). The linear regression model may not be appropriate in such cases.

Bernoulli trials

The econometrician observes $\{Y_i : i = 1, \ldots, n\}$, where the Y_i are iid random variables. The distribution of Y_i is given by

$$Y_i = \begin{cases} 1 & \text{with probability } \theta, \\ 0 & \text{with probability } 1-\theta. \end{cases}$$

This is the Bernoulli(θ) distribution, and

$$\begin{aligned} \mathrm{E}[Y_i] &= \theta \\ &= \Pr(Y_i = 1). \end{aligned} \tag{1}$$

The PMF of Y_i is

$$p(y_i, \theta) = \theta^{y_i} (1-\theta)^{1-y_i} \text{ for } y_i = 0, 1.$$

Thus, the log-likelihood function is given by

$$\begin{aligned} \log L_n(\theta) &= \left(n^{-1} \sum_{i=1}^{n} Y_i \right) \log \theta + \left(1 - n^{-1} \sum_{i=1}^{n} Y_i \right) \log(1-\theta) \\ &= \overline{Y}_n \log \theta + \left(1 - \overline{Y}_n \right) \log(1-\theta), \end{aligned}$$

where $\overline{Y}_n = n^{-1} \sum_{i=1}^{n} Y_i$. The ML estimator of θ is the sample mean:

$$\widehat{\theta}_n^{ML} = \overline{Y}_n.$$

Computing the second derivative,

$$\frac{d^2 \log p(Y_i, \theta)}{d\theta^2} = -\frac{Y_i}{\theta^2} - \frac{1-Y_i}{(1-\theta)^2}.$$

Equation (1) implies that

$$\begin{aligned} \mathrm{E}\left[\frac{d^2 \log p(Y_i, \theta)}{d\theta^2} \right] &= -\frac{1}{\theta} - \frac{1}{1-\theta} \\ &= -\frac{1}{\theta(1-\theta)}. \end{aligned}$$

Thus, in this model, the information (scalar) is given by

$$I(\theta) = \frac{1}{\theta(1-\theta)},$$

and the results in Lecture 14 imply that

$$\begin{aligned} \overline{Y}_n &\to_p \theta, \\ n^{1/2} \left(\overline{Y}_n - \theta \right) &\to_d N(0, \theta(1-\theta)). \end{aligned}$$

We can estimate the asymptotic variance consistently by $\overline{Y}_n\left(1-\overline{Y}_n\right)$. A $1-\alpha$ asymptotic confidence interval for θ can be constructed as follows:

$$\left[\overline{Y}_n \pm z_{1-\alpha/2}\sqrt{\frac{\overline{Y}_n\left(1-\overline{Y}_n\right)}{n}}\right].$$

The Bernoulli trials model is univariate. We now extend it to the case where the probability of Y_i taking the value 1 is a function of some exogenous explanatory variables.

Linear probability model

Suppose that the econometrician observes $\{(Y_i, X_i) : i = 1, \ldots, n\}$, where the Y_i are binary $\{0, 1\}$ variables, and the X_i are k-vectors of exogenous variables that explain $\Pr(Y_i = 1 \mid X_i)$. Let us assume that

$$\Pr(Y_i = 1 \mid X_i) = F\left(X_i^\top \beta\right),$$

for some function F. Unlike the Bernoulli trials model, the probability of Y_i taking the value one or zero is not constant and depends on some observable characteristics X_i.

Under the linear regression model,

$$\mathrm{E}[Y_i \mid X_i] = X_i^\top \beta,$$

then

$$F\left(X_i^\top \beta\right) = X_i^\top \beta.$$

That is, the probability of $Y_i = 1$ is a linear function of X_i. Such an assumption is a good starting point for the analysis, but it suffers from a serious flaw. If some of the components of X_i are continuous, $X_i^\top \beta$ cannot be restricted to the zero-one interval, and the model may yield probabilities outside $[0, 1]$. To avoid such predictions, we require F to map into $[0, 1]$, which forces F to be nonlinear. Any CDF is a natural choice for F.

Despite this limitation, the LPM remains useful in practice for estimating average partial effects, particularly when the main interest is in a treatment coefficient. Under correct specification of the conditional expectation, OLS provides a consistent estimator of the average marginal effect without requiring a distributional assumption on the errors.

Probit and logit models

It is convenient to introduce a latent (unobservable) variable Y_i^*, which can be interpreted as the net benefit or unobserved utility that determines the agent's binary decision. The usual linear regression model holds for Y_i^*:

$$Y_i^* = X_i^\top \beta + U_i.$$

We assume further that $\{(Y_i^*, X_i) : i = 1, \ldots, n\}$ are iid, and that

$$Y_i = \begin{cases} 1 & \text{if } Y_i^* > 0, \\ 0 & \text{if } Y_i^* \leq 0. \end{cases}$$

Let F be the conditional CDF of U_i given X_i:

$$F(u \mid X_i) = \Pr(U_i \leq u \mid X_i).$$

Then,

$$\begin{aligned} \Pr(Y_i = 1 \mid X_i) &= \Pr(Y_i^* > 0 \mid X_i) \\ &= \Pr\left(X_i^\top \beta + U_i > 0 \mid X_i\right) \\ &= \Pr\left(U_i > -X_i^\top \beta \mid X_i\right) \\ &= 1 - F\left(-X_i^\top \beta \mid X_i\right). \end{aligned}$$

Assume further that the distribution of U_i does not depend on X_i, so $F(u)$ replaces $F(u \mid X_i)$. The usual assumption in this framework is that F is symmetric around zero, that is,

$$F(u) = 1 - F(-u).$$

Under this assumption,

$$\Pr(Y_i = 1 \mid X_i) = F\left(X_i^\top \beta\right).$$

The common choices for F are

- Probit model: F is the standard normal CDF, denoted by Φ.
- Logit model: F is the logistic CDF, denoted by Λ:

$$\begin{aligned}\Lambda\left(X_i^\top \beta\right) &= \frac{\exp\left(X_i^\top \beta\right)}{\exp\left(X_i^\top \beta\right) + 1} \\ &= \frac{1}{1 + \exp\left(-X_i^\top \beta\right)}.\end{aligned}$$

For either choice of F, the model is estimated by maximum likelihood. The conditional PMF of Y_i given $X_i = x_i$ takes the same form as in the Bernoulli model, with θ replaced by $F\left(x_i^\top \beta\right)$:

$$p(y_i, \beta \mid x_i) = F\left(x_i^\top \beta\right)^{y_i}\left(1 - F\left(x_i^\top \beta\right)\right)^{1-y_i} \quad \text{for } y_i = 0, 1.$$

Thus, the log-likelihood is given by

$$\log L_n(\beta) = n^{-1}\sum_{i=1}^n Y_i \log F\left(X_i^\top \beta\right) + n^{-1}\sum_{i=1}^n (1 - Y_i)\log\left(1 - F\left(X_i^\top \beta\right)\right).$$

The first-order condition for $\widehat{\beta}_n^{ML}$ is

$$\begin{aligned}0 &= \frac{\partial \log L_n\left(\widehat{\beta}_n^{ML}\right)}{\partial \beta} \\ &= n^{-1}\sum_{i=1}^n Y_i \frac{\partial F\left(X_i^\top \widehat{\beta}_n^{ML}\right)/\partial \beta}{F\left(X_i^\top \widehat{\beta}_n^{ML}\right)} - n^{-1}\sum_{i=1}^n (1 - Y_i)\frac{\partial F\left(X_i^\top \widehat{\beta}_n^{ML}\right)/\partial \beta}{1 - F\left(X_i^\top \widehat{\beta}_n^{ML}\right)} \\ &= n^{-1}\sum_{i=1}^n Y_i \frac{f\left(X_i^\top \widehat{\beta}_n^{ML}\right)}{F\left(X_i^\top \widehat{\beta}_n^{ML}\right)} X_i - n^{-1}\sum_{i=1}^n (1 - Y_i)\frac{f\left(X_i^\top \widehat{\beta}_n^{ML}\right)}{1 - F\left(X_i^\top \widehat{\beta}_n^{ML}\right)} X_i,\end{aligned}$$

where f is the density corresponding to F:

$$f(u) = \frac{dF(u)}{du}.$$

No closed-form expression exists for $\widehat{\beta}_n^{ML}$; it must be computed numerically. Standard statistical software produces the ML estimates and asymptotic standard errors for both models.

For the logit model, the derivative of Λ satisfies

$$\frac{d\Lambda(u)}{du} = \Lambda(u)\left(1 - \Lambda(u)\right). \tag{2}$$

Therefore, for the logit model,

$$\frac{f\left(X_i^\top\beta\right)}{F\left(X_i^\top\beta\right)} = 1-\Lambda\left(X_i^\top\beta\right),$$
$$\frac{f\left(X_i^\top\beta\right)}{1-F\left(X_i^\top\beta\right)} = \Lambda\left(X_i^\top\beta\right).$$

The first-order condition simplifies to

$$\begin{aligned}\frac{\partial \log L_n\left(\widehat{\beta}_n^{ML}\right)}{\partial\beta} &= n^{-1}\sum_{i=1}^{n}\left(Y_i-\Lambda\left(X_i^\top\widehat{\beta}_n^{ML}\right)\right)X_i \\ &= 0.\end{aligned}$$

Similarly,

$$\frac{\partial \log p\left(y_i,\beta \mid x_i\right)}{\partial\beta} = \left(y_i-\Lambda\left(x_i^\top\beta\right)\right)x_i,$$

and

$$\frac{\partial^2 \log p\left(y_i,\beta \mid x_i\right)}{\partial\beta\partial\beta^\top} = -\Lambda\left(x_i^\top\beta\right)\left(1-\Lambda\left(x_i^\top\beta\right)\right)x_i x_i^\top.$$

Thus, the information matrix is given by

$$\begin{aligned}I\left(\beta\right) &= -\operatorname{E}\left[\frac{\partial^2 \log p\left(Y_i,\beta \mid X_i\right)}{\partial\beta\partial\beta^\top}\right] \\ &= \operatorname{E}\left[\Lambda\left(X_i^\top\beta\right)\left(1-\Lambda\left(X_i^\top\beta\right)\right)X_i X_i^\top\right].\end{aligned}$$

Hence, for the logit model, $\widehat{\beta}_n^{ML}\to_p\beta$, and

$$n^{1/2}\left(\widehat{\beta}_n^{ML}-\beta\right)\to_d N\left(0,\left(\operatorname{E}\left[\Lambda\left(X_i^\top\beta\right)\left(1-\Lambda\left(X_i^\top\beta\right)\right)X_i X_i^\top\right]\right)^{-1}\right).$$

The asymptotic variance-covariance matrix of $\widehat{\beta}_n^{ML}$ can be estimated consistently by

$$\left(n^{-1}\sum_{i=1}^{n}\Lambda\left(X_i^\top\widehat{\beta}_n^{ML}\right)\left(1-\Lambda\left(X_i^\top\widehat{\beta}_n^{ML}\right)\right)X_i X_i^\top\right)^{-1}.$$

Marginal effects

In the linear regression model, the marginal effect of X_i on $\operatorname{E}\left[Y_i \mid X_i\right]$ is β. In nonlinear binary choice models such as probit or logit,

$$\begin{aligned}\frac{\partial \operatorname{E}\left[Y_i \mid X_i\right]}{\partial X_i} &= \frac{\partial \Pr\left(Y_i=1 \mid X_i\right)}{\partial X_i} \\ &= \frac{\partial F\left(X_i^\top\beta\right)}{\partial X_i} \\ &= f\left(X_i^\top\beta\right)\beta,\end{aligned}$$

and

$$\begin{aligned}\frac{\partial \Pr\left(Y_i=0 \mid X_i\right)}{\partial X_i} &= \frac{\partial\left(1-\Pr\left(Y_i=1 \mid X_i\right)\right)}{\partial X_i} \\ &= -f\left(X_i^\top\beta\right)\beta.\end{aligned}$$

Thus, in the case of the probit model, the marginal effect of X_i on $\Pr(Y_i = 1 \mid X_i)$ is given by

$$\phi\left(X_i^\top \beta\right)\beta,$$

where $\phi(u) = d\Phi(u)/du$ is the standard normal density. In the case of logit, equation (2) implies that $\partial \Pr(Y_i = 1 \mid X_i)/\partial X_i$ is given by

$$\Lambda(X_i^\top \beta)\left(1 - \Lambda(X_i^\top \beta)\right)\beta.$$

The marginal effects can be estimated by replacing the unknown coefficients β with their ML estimators. For a fixed $X_i = x$,

$$\frac{\partial \widehat{\Pr(Y_i = 1 \mid X_i = x)}}{\partial X_i} = f\left(x^\top \widehat{\beta}_n^{ML}\right)\widehat{\beta}_n^{ML}.$$

By Slutsky's theorem and the consistency of $\widehat{\beta}_n^{ML}$,

$$\begin{aligned}
\frac{\partial \widehat{\Pr(Y_i = 1 \mid X_i)}}{\partial X_i} &= f\left(x^\top \widehat{\beta}_n^{ML}\right)\widehat{\beta}_n^{ML} \\
&\to_p f\left(x^\top \beta\right)\beta \\
&= \frac{\partial \Pr(Y_i = 1 \mid x)}{\partial X_i},
\end{aligned}$$

provided that f is continuous.

The asymptotic distribution of $f\left(x^\top \widehat{\beta}_n^{ML}\right)\widehat{\beta}_n^{ML}$ can be obtained using the delta method:

$$\begin{aligned}
n^{1/2}\left(f\left(x^\top \widehat{\beta}_n^{ML}\right)\widehat{\beta}_n^{ML} - f\left(x^\top \beta\right)\beta\right) &\to_d N\left(0, \frac{\partial\left(f\left(x^\top \beta\right)\beta\right)}{\partial \beta^\top} I^{-1}(\beta)\frac{\partial\left(f\left(x^\top \beta\right)\beta^\top\right)}{\partial \beta}\right) \\
&= N\left(0, \left(f'\left(x^\top \beta\right)\beta x^\top + f\left(x^\top \beta\right)I_k\right)I^{-1}(\beta)\left(f'\left(x^\top \beta\right)\beta x^\top + f\left(x^\top \beta\right)I_k\right)^\top\right).
\end{aligned}$$